

Assignment 4

You are given a map of 23 cities (named as A, B, C, D, ..., X, Y, Z respectively). The information of this map is available in the file "city 1.txt" in the same folder. Every road connection is represented by a row in the table as illustrated below, where the first and the second columns correspond to the two cities that are directly connected and the third column denotes the distance of the road that connects the two cities.

Location ID 1	Location ID 2	Distance
A	B	2
A	E	2
A	W	1
...	...	...

Your assignment in this lab is to write a computer program based on the principle of dynamic programming to implement the following two functions:

- 1) Calculate the optimal values of all cities given that the destination is city F
- 2) Find the shortest path from each city to F.

In your lab report you need to, as exercise, solve a small problem (as given in Figure 1) with manual calculation. The purpose of this exercise is to demonstrate your understanding of the idea and principle. Assume that the destination in this small problem is node E. Please write out the results of all the iterative steps to acquire the optimal values of nodes. You also need to show how you can find the shortest path to E from each of the other nodes.

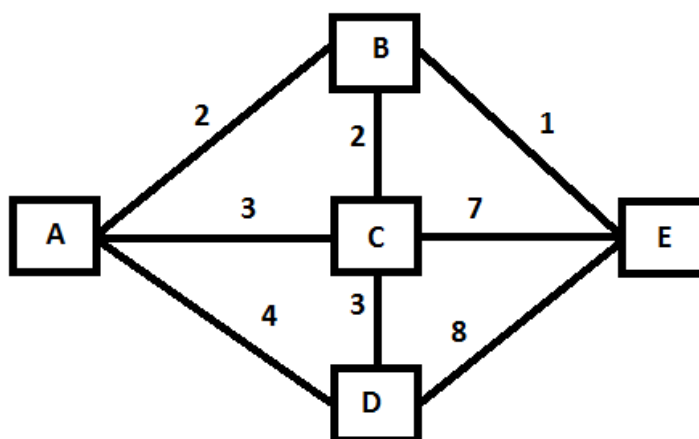


Figure 1: The graph for a small problem

Before submitting the report, you should present your program and results to Ning Xiong. After that, and only if your work is approved, you can then submit the report in CANVAS.